

Tangential Phase Error as a Precursor to Catastrophic Channel Escape

1 Motivation

Some rollout predictions eventually leave the physically admissible microfluidic channel. The purpose of this analysis is to test whether these catastrophic channel escapes are associated with accumulated longitudinal phase error along the channel trajectory.

Catastrophic escape is defined using the existing channel-admissibility criterion:

$$\text{outside_fraction} \geq 0.95$$

at any evaluable rollout step. Each rollout trajectory is therefore grouped as either escaping or non-escaping. This grouping is based on the predicted trajectory over the rollout horizon, while the signed phase error is measured relative to the corresponding true trajectory.

2 Signed Tangential Error

For droplet instance i at rollout step k , the signed tangential position error is

$$e_{\text{tan},i}(k) = (\hat{\mathbf{x}}_i(k) - \mathbf{x}_i(k)) \cdot \mathbf{t}_i(k),$$

where $\hat{\mathbf{x}}_i(k)$ is the predicted droplet position, $\mathbf{x}_i(k)$ is the true position, and $\mathbf{t}_i(k)$ is the consistently oriented local channel tangent evaluated at the true position.

The implementation defines the signed error as predicted position minus true position, dotted with the true-position tangent. Therefore, positive e_{tan} means the predicted droplet is ahead of the true droplet along the oriented channel tangent, while negative e_{tan} means the predicted droplet lags behind the true droplet.

At each rollout step, the plotted statistic is the median signed tangential error over all valid rollout instances in each trajectory class. Because the validation set contains overlapping rollout windows, these summaries should be interpreted as statistics over valid rollout instances, not necessarily over unique physical track IDs.

3 Result

Figure 1 shows a clear separation between escaping and non-escaping rollout trajectories. Non-escaping trajectories remain relatively close to zero signed tangential error, while escaping trajectories develop a substantially larger systematic signed tangential error over rollout time. This separation is visible across all three model classes.

The result indicates that catastrophic channel escapes are not merely isolated geometric violations at the final moment of failure. They are strongly associated with an earlier and progressively

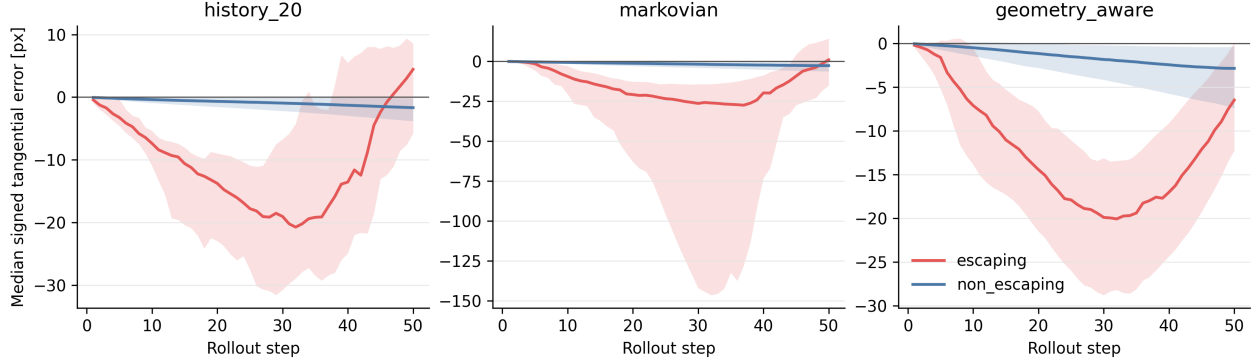


Figure 1: Stepwise signed tangential error for catastrophic escaping and non-escaping rollout trajectories. Each panel corresponds to one model class: `history_20`, `markovian`, and `geometry_aware`. Curves show the median signed tangential error at each rollout step, and the shaded bands show the interquartile range from the 25th to 75th percentiles. Negative values indicate predictions lagging behind the true droplet along the true-position channel tangent; positive values indicate predictions ahead of the true droplet.

accumulating longitudinal phase mismatch between predicted and true droplet motion. This does not prove that phase error is the sole cause of escape, but it is consistent with phase error acting as a precursor that likely contributes to later geometric failure.

4 Physical Interpretation

A plausible physical mechanism is as follows. First, the model inaccurately predicts the amount or duration of local slowing. Second, the predicted droplet becomes out of phase with the true droplet along the channel. Third, subsequent predicted motion becomes inappropriate for the droplet’s current predicted spatial location. Finally, the trajectory can cross the channel wall and undergo catastrophic escape.

This phase error can originate from incorrectly learned junction dynamics. At a junction, the real droplet may slow or stall because of hydrodynamic interactions and local flow redistribution. If the model does not capture the correct magnitude or duration of this slowdown, it accumulates longitudinal phase error. The droplet may then enter subsequent curvature or branch dynamics from the wrong spatial phase, increasing the risk of wall escape.

Phase mismatch may occur in either direction: the predicted droplet may lag behind the truth, or it may move ahead of the truth. Both cases can make later predicted motion inappropriate for the predicted location, particularly near junctions, bends, and crowded interaction regions.

5 Implications

Geometry regularization alone cannot fully correct this failure mode if it only penalizes the eventual geometric violation. Such a penalty can discourage wall crossing, but it does not necessarily restore the missing junction or interaction dynamics that created the accumulated phase error.

The finding motivates a more physics-informed model state. A natural next step is to include a position-dependent background fluid-flow representation, with later extensions potentially incorporating droplet morphology or deformation descriptors. This should be viewed as motivation for future work, not as a demonstrated solution in the present analysis.

6 Limitations

This analysis is associative: trajectories are grouped by whether they eventually escape, and the observed separation does not establish a unique causal mechanism. The analysis also does not establish the exact source of the predicted velocity direction, nor does it prove that phase error is the only mechanism of catastrophic escape. Finally, because overlapping rollout windows are included, the summary statistics may reflect repeated rollout instances from the same physical droplet track.